

JanGyan — Learn Without Limits

Category : Cyber Safety & Digital Security

Module 8 - Safe Internet Usage

Module Overview

This handbook is your complete, beginner-to-advanced guide to **Safe Internet Usage** — covering public Wi-Fi, downloads, browser safety, and fake websites. These four everyday habits shape almost every risk you face simply by being online, and small adjustments to each can dramatically reduce your exposure to fraud and malware.

By the end of this module, you will not need to search anywhere else to browse, download, and connect safely.

Learning Objectives

After completing this module, you will be able to:

- Understand the real risks of using public Wi-Fi and how to reduce them
- Download files and apps safely, avoiding common malware traps
- Use your browser's built-in safety features effectively
- Recognise fake and look-alike websites before entering any information
- Verify a website's authenticity using simple, reliable checks
- Know exactly where and how to report a fake or malicious website in India

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Why This Topic Matters

Google's Safe Browsing technology now protects over **5 billion devices every day**, and its Enhanced Protection mode — used by more than a billion Chrome users — makes people roughly **twice as safe** from phishing and scam websites compared to standard protection. That scale reflects just how constant the threat of fake and malicious websites has become.

> *Reality Check: Surveys consistently find that most people express concern about public Wi-Fi safety, yet a large share still access banking apps, email, or social media on unsecured public networks without taking any extra precautions.*

The internet doesn't warn you before a network is unsafe, a download is malicious, or a website is fake — recognising the signs yourself is the only real protection.

Introduction

Safe internet usage covers the everyday choices that determine how exposed you are to fraud and malware:

- **Public Wi-Fi:** Free wireless networks in cafes, airports, malls, and stations, which vary widely in security
- **Downloads:** Files and apps you install onto your device, some of which may carry hidden malware
- **Browser Safety:** The settings and habits that protect you while you browse

- **Fake Websites:** Look-alike or fabricated sites designed to steal information or money

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Fundamental Concepts

HTTP vs HTTPS

Protocol	Meaning	Safety Signal
HTTP	Data sent in plain, readable form	Avoid entering sensitive information
HTTPS	Data encrypted in transit (padlock icon in browser)	Safer, but does not guarantee the site itself is genuine

HTTPS confirms your connection to the site is encrypted — it does **not** confirm the site is trustworthy. Fake websites can and often do use HTTPS too.

Anatomy of a Web Address

`https://www.examplebank.com/login`

- **Domain:** `examplebank.com` — the part that identifies the real owner
- **Subdomain:** `www.` — a prefix, easily faked or altered
- **Path:** `/login` — the specific page, irrelevant to verifying authenticity

Always check the **exact domain name** carefully — fake sites often use extra words, hyphens, or subtly misspelled versions (e.g., `examplebank-secure-login.com` instead of `examplebank.com`).

Complete Detailed Explanation

Part A — Public Wi-Fi Safety

How attacks happen:

- **Unencrypted Networks:** Many public Wi-Fi networks don't encrypt traffic, letting anyone nearby with the right tools potentially read what you send.
- **Evil Twin Networks:** A fake hotspot with a name nearly identical to a genuine one (e.g., "Airport_Free_WiFi" vs "Airport_FreeWiFi"), created specifically to intercept connecting users' data.
- **Session Hijacking:** An attacker steals your active login session, gaining account access without ever needing your password.
- **Malware Injection:** Some compromised or malicious networks push malicious pop-ups or redirect downloads to install malware silently.

How to reduce risk: Avoid banking, UPI, or sensitive logins on public Wi-Fi; verify the exact network name with staff; keep auto-connect disabled; and use a trusted VPN if you must handle sensitive tasks on public networks.

Part B — Downloads

- **Official Sources Only:** Download apps from official stores and software directly from the publisher's verified website, not third-party download aggregators.
- **File Extensions Matter:** Be cautious of unexpected executable files (`.exe`, `.apk`, `.bat`) received via email, chat, or unfamiliar sites.
- **Pirated/Cracked Software Risk:** "Free" versions of paid software are a common way malware is distributed, since they require disabling security checks to install.
- **Browser Download Warnings:** Take seriously any warning your browser shows about a download being unverified or potentially harmful — don't dismiss it automatically.

Part C — Browser Safety

- **Google Safe Browsing:** A background protection system (used by Chrome, Firefox, and Safari) that checks websites and downloads against a constantly updated list of known dangerous sites, protecting over 5 billion devices daily.
- **Enhanced Protection Mode:** An optional, stronger setting in Chrome using AI to proactively detect new phishing and scam sites in real time, rather than relying only on a known-bad list.
- **Browser Extensions:** Only install extensions from official browser stores, and review their permissions — a malicious extension can read everything you type or view.
- **Incognito/Private Browsing:** Prevents your browsing history from being saved locally, but does **not** make you anonymous online or protect you from phishing/malware.

Part D — Fake Websites

Common tactics:

- **Typosquatting:** Registering domains with common misspellings of popular sites (e.g., a single swapped letter in a bank's name).
- **Look-Alike Government Portals:** Fake sites mimicking official portals (PAN card, passport, train ticket booking) that charge unnecessary "service fees" or steal personal/payment details.
- **Fake Shopping/Deal Sites:** Websites offering unrealistically large discounts, designed purely to capture payment information without ever shipping a product.

How to verify a website is genuine: Check the exact domain spelling, look for HTTPS as a baseline (not a guarantee), search independently for the organisation's official website rather than trusting a link, and be sceptical of any deal that feels too good to be true.

Important Terminologies

Term	Simple Meaning
HTTPS	An encrypted connection between your browser and a website
Domain	The core identifying part of a website's address
Typosquatting	Registering a fake domain based on a common misspelling of a real one
Evil Twin Network	A fake Wi-Fi hotspot mimicking a genuine one to intercept data
Session Hijacking	Stealing an active login session to access an account without a password
VPN (Virtual Private Network)	A tool that encrypts your internet traffic for extra privacy and security
Safe Browsing	Google's system for detecting and warning users about dangerous websites
Browser Extension	An add-on that adds functionality to a browser, with its own permissions

Step-by-Step Understanding

How to Safely Use Public Wi-Fi

1. Verify the exact network name with staff before connecting
2. Turn off auto-connect for Wi-Fi on your device
3. Avoid banking, UPI, or other sensitive logins while connected
4. Use a trusted VPN if you must handle anything sensitive
5. Forget the network on your device after you're done using it

How to Verify a Website Is Genuine

1. Check the exact domain spelling character by character
2. Look for HTTPS as a baseline signal (not proof of legitimacy)
3. Independently search for the organisation's official website rather than clicking a provided link
4. Be sceptical of unrealistic discounts, urgent deadlines, or unusual payment requests
5. When in doubt, contact the organisation directly through a verified channel to confirm

Real-Life Indian Examples

(Widely reported public patterns, described for educational awareness.)

- **Fake Government Portal Look-Alikes:** Fraudulent websites mimicking official portals for PAN card applications, passport services, and train ticket booking have repeatedly been flagged by Indian authorities for charging unnecessary fees or harvesting personal data.

- **Fake Festive-Season Shopping Sites:** Security researchers have observed recurring spikes in fake e-commerce websites offering steep "festival discounts," particularly around major Indian shopping seasons, many of which never deliver purchased goods.
- **Public Wi-Fi Fraud at Transit Hubs:** Reports of fraudulent or unsecured hotspots at airports, railway stations, and malls in India highlight ongoing risks for travellers who connect to convenience networks without verification.

Practical Scenarios

Scenario 1: You're at an airport and see two very similarly named free Wi-Fi networks.
Ask airport staff which network is genuine before connecting to either — this is a classic evil-twin setup.

Scenario 2: A website offering a government service asks for an unusually high "processing fee" before you can proceed.
Verify the service directly on the officially known government domain — many look-alike sites add unnecessary fees.

Scenario 3: Your browser warns that a file you're about to download is not commonly downloaded and could be harmful.
Don't dismiss this warning automatically — verify the source independently before proceeding.

Scenario 4: You need to check your bank balance while connected to a café's public Wi-Fi.
Switch to your mobile data instead, or use a trusted VPN, rather than doing banking directly over public Wi-Fi.

Government Rules / Guidelines

- **CERT-In:** Issues advisories on fake websites, phishing domains, and safe browsing practices, and coordinates with global platforms to flag malicious sites affecting Indian users.
- **MeitY:** Oversees India's broader digital safety framework, including guidance around safe internet usage for citizens.
- **Google Safe Browsing:** While not a government body, its detection lists are integrated into most major browsers used in India, forming a critical layer of everyday protection.

Laws & Policies

Law / Provision	What It Covers
IT Act, 2000 — Section 66D	Punishes cheating by impersonation using a computer resource — the core offence behind most fake websites
IT Act, 2000 — Section 66C	Punishes identity theft, often the goal of fake website data harvesting
IT Act, 2000 — Section 43	Civil liability for unauthorised access or damage relating to computer systems, applicable to malware from malicious downloads

Important Statistics

Metric	Figure
Devices protected daily by Google Safe Browsing	Over 5 billion
Chrome users on Enhanced Protection mode	Over 1 billion
Relative phishing/scam protection with Enhanced Protection	About 2x safer than Standard Protection
Suspicious files scanned monthly by Chrome's Enhanced Protection	Over 300,000
Public Wi-Fi hotspots deployed globally (2025)	Approximately 950 million

(Figures are approximate, drawn from Google's published transparency and security reports through 2025–2026.)

Best Practices

- Avoid banking, UPI, or other sensitive logins on public Wi-Fi
- Verify exact Wi-Fi network names with staff before connecting
- Download apps and software only from official, verified sources
- Enable your browser's Enhanced/Safe Browsing protection
- Check domain spelling carefully before entering any information on a website
- Use a trusted VPN when you must handle sensitive tasks on public networks

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Common Mistakes

- Connecting to public Wi-Fi networks without verifying their name
- Doing banking or UPI transactions over unsecured public Wi-Fi
- Downloading pirated or cracked software from unofficial sites
- Ignoring browser warnings about unsafe downloads or websites
- Assuming HTTPS alone means a website is genuine
- Trusting a link instead of independently searching for an organisation's official site

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Myths vs Facts

Myth	Fact
"HTTPS means a website is definitely safe and genuine"	HTTPS only means the connection is encrypted — fake websites can use HTTPS too
"Incognito/private browsing makes me anonymous online"	It only avoids saving local browsing history — it doesn't hide you from websites or protect against phishing
"Public Wi-Fi at well-known places is always safe"	Even reputable venues can have compromised networks or nearby evil-twin hotspots
"If a website looks professional, it must be genuine"	Fake websites can closely copy a genuine site's design and branding
"Free VPNs offer the same protection as trusted paid ones"	Some free VPNs have weak security or may log and sell user data themselves

Do's and Don'ts

Do	Don't
Verify Wi-Fi network names before connecting	Don't connect to open networks without checking with staff
Download apps/software only from official sources	Don't install pirated or cracked software from unofficial sites
Check domain spelling carefully before entering data	Don't assume a professional-looking site is automatically genuine
Use a trusted VPN for sensitive tasks on public Wi-Fi	Don't do banking/UPI transactions on unsecured public networks
Take browser security warnings seriously	Don't dismiss "unsafe site/download" warnings without checking further

Safety Tips

For Public Wi-Fi

- Turn off auto-connect and forget networks after use
- Avoid sensitive logins; use mobile data or a VPN instead
- Verify exact network names with venue staff

For Downloads

- Stick to official app stores and verified publisher websites
- Avoid pirated/cracked software entirely
- Pay attention to unexpected file types (.exe, .apk) from unfamiliar sources

For Browsing

- Enable Enhanced/Safe Browsing protection in your browser
- Only install browser extensions from official stores, and review their permissions
- Don't rely on incognito/private mode for actual security

For Websites

- Check domain spelling character by character
- Search independently for official sites rather than clicking provided links
- Be sceptical of unrealistic discounts or urgent "pay now" demands

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Practical Checklist

- ☐ Learned to verify Wi-Fi network names before connecting
- ☐ Avoid banking/UPI on public Wi-Fi without a VPN
- ☐ Download apps/software only from official sources
- ☐ Enabled Enhanced/Safe Browsing protection in my browser
- ☐ Know to check domain spelling carefully before entering data
- ☐ Understand incognito mode doesn't provide real security
- ☐ Saved the 1930 cybercrime helpline for reporting fake websites

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Quick Revision Notes

- HTTPS means encrypted connection, not proof of a genuine website
- Public Wi-Fi risks include evil twins, session hijacking, and malware injection
- Download only from official app stores/verified publisher sites
- Google Safe Browsing protects over 5 billion devices daily; Enhanced Protection adds AI-based real-time detection
- Fake websites often rely on typosquatting or copying government/brand look
- Always verify sites independently rather than trusting a provided link

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Key Takeaways

1. A padlock icon (HTTPS) means encryption, not trustworthiness
2. Public Wi-Fi convenience comes with real, well-documented risks — plan around them
3. Official sources for downloads and browser extensions matter more than convenience
4. Fake websites succeed by copying appearances — verify domains, not designs
5. Simple habits (checking domain names, avoiding sensitive logins on public Wi-Fi) prevent most of these risks

Glossary

- **Domain** — The core identifying part of a website's address
- **Evil Twin Network** — A fake Wi-Fi hotspot mimicking a genuine one
- **HTTPS** — An encrypted connection between browser and website
- **Incognito/Private Browsing** — A mode that avoids saving local browsing history only
- **Session Hijacking** — Stealing an active login session to access an account
- **Typosquatting** — Registering a fake domain based on a common misspelling
- **VPN** — A tool encrypting internet traffic for privacy and security

Frequently Asked Questions

1. Is public Wi-Fi ever safe to use?
It can be used for non-sensitive browsing with precautions, but sensitive logins like banking should be avoided or done via VPN.

2. Does HTTPS guarantee a website is genuine?
No — it only confirms the connection is encrypted; fake websites can also use HTTPS.

3. What is typosquatting?
Registering a fake website domain based on a common misspelling of a genuine one.

4. Is incognito/private browsing mode a security feature?
Not really — it only prevents local browsing history from being saved; it doesn't protect against phishing or malware.

5. What is an evil twin Wi-Fi network?
A fake hotspot with a name nearly identical to a genuine one, created to intercept users' data.

6. Is it safe to download cracked or pirated software?
No — this is a common way malware is distributed, since it requires bypassing normal security checks.

7. What is Google Safe Browsing?
A protection system used by major browsers that checks websites/downloads against a list of known dangerous sites.

8. What is Chrome's Enhanced Protection mode?
An optional setting using AI to proactively detect new phishing and scam sites in real time.

9. How can I tell if a website is fake?
Check the exact domain spelling, be sceptical of unrealistic offers, and verify independently rather than trusting a link.

10. Should I do UPI transactions on public Wi-Fi?
It's best avoided; use mobile data or a trusted VPN instead.

11. What is session hijacking?
An attacker stealing your active login session to access your account without needing your password.

12. Are all browser extensions safe to install?
No — only install extensions from official browser stores and review their requested permissions.

13. What should I do if my browser warns a download is unsafe?
Take the warning seriously and verify the source independently before proceeding.

14. How do fake government portal websites usually operate?
They mimic official portals closely and often charge unnecessary "fees" or harvest personal/payment details.

15. What is a VPN, and does it make public Wi-Fi fully safe?
A VPN encrypts your traffic, significantly reducing (but not eliminating) risk on public Wi-Fi.

16. Can a professional-looking website still be fake?
Yes — fake websites frequently copy the design and branding of genuine ones closely.

17. What's the safest way to reach a government service website?
Search for and navigate directly to the officially known domain rather than clicking a shared link.

18. Should I trust free Wi-Fi with a very generic name like "Free_WiFi"?
Be cautious — verify with venue staff, since generic names are easy for attackers to imitate.

19. Is it safe to auto-connect to known Wi-Fi networks?
It's safer to disable auto-connect, since attacker-controlled networks can sometimes mimic saved network names.

20. What should I check before entering payment details on any website?
Verify the exact domain spelling, look for HTTPS as a baseline, and confirm the site independently if unsure.

21. Are unrealistic discount offers on shopping sites a red flag?
Yes — extremely steep discounts are a common tactic used by fake shopping websites.

22. Where can I report a fake website in India?
Through the National Cyber Crime Reporting Portal (cybercrime.gov.in) or the 1930 helpline.

Suggested Infographics, Flowcharts & Illustrations

(For JanGyan design team — placement suggestions only; no images generated here)

- **"Anatomy of a Web Address" annotated diagram** → within "Fundamental Concepts"
- **Public Wi-Fi risk icons (evil twin, session hijacking, malware injection)** → within "Complete Detailed Explanation, Part A"
- **Flowchart: "Is This Website Genuine?"** → within "Step-by-Step Understanding"
- **HTTPS padlock myth-buster visual card** → within "Myths vs Facts"
- **"Never Bank on Public Wi-Fi" warning card** → near "Practical Checklist"

Official References

- Google Safe Browsing — Transparency Report and Enhanced Protection documentation
- Indian Computer Emergency Response Team (CERT-In) — advisories on fake websites and phishing domains
- Information Technology Act, 2000 (as amended), India Code portal
- Ministry of Electronics and Information Technology (MeitY) — citizen digital safety guidance

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Official Government Website Links

- CERT-In: **https://www.cert-in.org.in**
- MeitY: **https://www.meity.gov.in**
- National Cyber Crime Reporting Portal: **https://cybercrime.gov.in**
- Digital India: **https://www.digitalindia.gov.in**

Cybercrime Helpline (24×7, Toll-Free): 1930

Conclusion

Safe internet usage isn't about avoiding the internet's conveniences — it's about a handful of quick checks: verifying a Wi-Fi network's name, downloading only from official sources, trusting your browser's warnings, and checking a domain letter by letter before typing in anything sensitive. These small habits, repeated consistently, close off the vast majority of everyday online risks.

This is only the next step in your Cyber Safety journey with JanGyan. Stay alert, stay informed, and remember: **verify the network, verify the source, verify the domain — every single time.**

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Module 8 of the Cyber Safety & Digital Security Category